Week 7 Updates

New architecture

- Maximally transmit time embedding by keeping separate from the global jet feature embedding
- Only use lorentz-invariant jet features (# constituents, one-hot encoded type) to produce the global embedding
- Allow reweighting of current particle in linear combination based on messages
- Follow convention of E(3) GNNs (Satorras, Hoogeboom, and Welling 2022) for scaling of particles in linear combination

New architecture details I

x = Cartesian particle 4-momenta, $\phi.$ = learnable functions,
$\psi(\cdot) = \text{sgn}(\cdot)\log(|\cdot|+1)$,$N$ = # particles, $||\cdot||, \langle\cdot\rangle$ = Minkowski norm and inner product

1. Embed scalar time value to random Fourier features through two fully cnnected layers with 32 and 64 nodes to producd t_{emb}
2. Pass one-hot encoded jet type j_t and number of constituents N through two fully-connected layers with 32 and 64 nodes to produce global embedding g^0
3. Initialize particle cloud $x_1 \dots x_N$, and zero-pad up to 150 if necessary
4. Stack L Lorentz-group equivariant layers
   4.1 Input to the $l+1$-th layer is t_{emb}, g^0, $g^L, x_1 \cdots x_N$
   4.2 Calculate messages $m_i j^l = \phi_e^l(\psi(||x_i^l - x_j^l||^2), \psi(\langle x_i, x_j\rangle), g^0, g^l, t_{\text{emb}})$
   4.3 Cache the message aggregation

$$\sum_{i,j \in N} \phi_m^l(m_{ij}^l) m_{ij}$$

New architecture details II

4.4 Update the global embedding

$$g^{l+1} = \phi_g^l\left(g^0, g^l, t_{\mathsf{emb}}\frac{\alpha_l}{N^2}\sum_{i,j\in N}\phi_m^l(m_{ij}^l)m_{ij}\right)$$

where ϕ_m^l outputs a scalar and α_l is a scalar

4.5 Apply a linear combination to update the 4-vectors

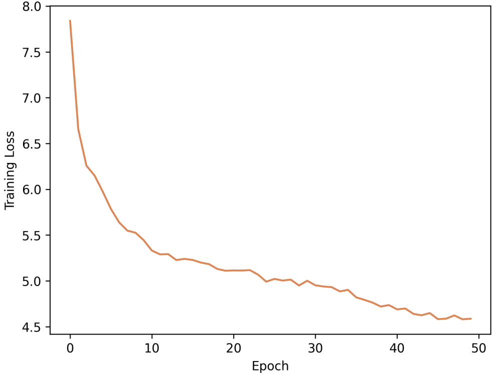
$$x_i^{l+1} = x_i^l + \gamma^l \sum_{j\neq i}\phi_x^l(g^0, g^l, t_{\mathsf{emb}}, m_{ij})\frac{x_i^l - x_j^l}{1 + ||x_i^l - x_j^l||}$$

where ϕ_w^l and ϕ_x^l output scalars

4.6 Final output $v_\theta(t, x^0)_i = x_i^L$

Video of output

Training loss



Plot suggests probably need ≥ 500 epochs for FM convergence

Conclusions

Positives:

- Time dependence is clearly present
- LorentzNet seems sufficiently expressive: the output of the model is nonzero (even if it *points* to $\mathbf{0}$)
- Outliers most likely stem from insufficient training data coverage - seems fixable

Hypotheses for performance:

1. **Bug in zero padding leading to collapse to $\mathbf{0}$**
2. Diffusion collapse to approximate centre of data space $\mathbf{0}$

Experiments for debugging

1. Train on gluons only (ongoing, 44/50 epochs)
2. Artificially translate training data away from $\mathbf{0}$ to (a, a, a, a) where $a \approx \mathcal{O}(10)$
    2.1 If the model still maps to $\mathbf{0}$: likely a zero padding issue
    2.2 If the model maps to (a, a, a, a): likely a diffusion collapse issue

Remaining work (if the collapse issue is resolved)

Core experimental work:

1. Ablation over training objectives (FM OT, variance preserving, variance exploding)
2. Likelihood estimation of output - boilerplate implementation

Potentially achievable novel contributions (to jet generation) in 2 weeks:

1. Classifier-free guidance on jet type and constituents (Ho and Salimans 2022; Fan et al. 2025)
2. Mean flows for one-shot generation (Geng et al. 2025)
3. Interval conditioning for ‘cuts’ during inference (Bansal et al. 2023; Gloeckler et al. 2024)

References I

Satorras, Victor Garcia, Emiel Hoogeboom, and Max Welling (Feb. 2022). *E(n) Equivariant Graph Neural Networks*. DOI: 10.48550/arXiv.2102.09844. arXiv: 2102.09844 [cs]. (Visited on 06/25/2025).

Ho, Jonathan and Tim Salimans (July 26, 2022). *Classifier-Free Diffusion Guidance*. DOI: 10.48550/arXiv.2207.12598. arXiv: 2207.12598 [cs]. URL: http://arxiv.org/abs/2207.12598 (visited on 08/05/2025). Pre-published.

Fan, Weichen et al. (Apr. 2025). *CFG-Zero*: Improved Classifier-Free Guidance for Flow Matching Models*. DOI: 10.48550/arXiv.2503.18886. arXiv: 2503.18886 [cs]. (Visited on 08/05/2025).

Geng, Zhengyang et al. (May 2025). *Mean Flows for One-step Generative Modeling*. DOI: 10.48550/arXiv.2505.13447. arXiv: 2505.13447 [cs]. (Visited on 06/24/2025).

References II

Bansal, Arpit et al. (Feb. 14, 2023). *Universal Guidance for Diffusion Models*. DOI: 10.48550/arXiv.2302.07121. arXiv: 2302.07121 [cs]. URL: http://arxiv.org/abs/2302.07121 (visited on 08/05/2025). Pre-published.

Gloeckler, Manuel et al. (July 2024). *All-in-One Simulation-Based Inference*. DOI: 10.48550/arXiv.2404.09636. arXiv: 2404.09636 [cs]. (Visited on 06/26/2025).

Appendices

Interval conditioning

- Guided diffusion: with context $\mathbf{y}$, $s(\hat{\mathbf{x}}_t, t|\mathbf{y}) \approx (\hat{\mathbf{x}}_t, t) + \nabla_{\hat{\mathbf{x}}_t} \log p_t(\mathbf{y}|\hat{\mathbf{x}}_t)$
- Bansal et al. 2023 show diffusion models can be guided by arbitrary functions
- Gloeckler et al. 2024 propose using $s_\phi(\hat{\mathbf{x}}_t, t|c) \approx s_\phi(\hat{\mathbf{x}}_t, t) + \nabla_{\hat{\mathbf{x}}_t} \log \sigma(-s(t)c(\hat{\mathbf{x}}_t))$ where
  - σ is the sigmoid function
  - $s(t)$ is a scalar function s.t. $s(t) \to \infty$ as $t \to 1$ (where $t = 1$ is data)
  - c is a constraint function, e.g., to enforce $\hat{\mathbf{x}} < u$, let $c(\hat{\mathbf{x}}) = \hat{\mathbf{x}} - u$

Will need to do semi-original work on checking if applicable to / generalizing this to flow matching